

PPG based AFib Prediction at Resource Constrained Environments

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Abstract—Paroxysmal atrial fibrillation (PAF) is episodic and often asymptomatic, so it frequently escapes the brief window of a clinical visit; anticipating its onset from a wearable stream would open a window for early intervention. We present a patient-aware early-warning scheme for atrial fibrillation (AF) built on RR-interval (RRI) sequences, the compact heart-rhythm signal that smartwatches and single-lead patches derive natively from either the electrocardiogram or photoplethysmography. A deliberately small multi-branch convolutional encoder is pretrained with a *patient-aware contrastive objective* whose positive pairs are restricted to same-patient, same-class segments, so it preserves each individual’s own sinus-rhythm baseline rather than collapsing all patients into one shared prototype. On the strictly patient-independent IRIDIA-AF benchmark the frozen representation reaches an area under the receiver operating characteristic curve (AUROC) of 0.989 ± 0.003 with $2.6\times$ lower seed-to-seed variance than supervised contrastive learning. Reusing this frozen embedding without any retraining, a per-patient sinus-rhythm-to-AF projection yields a scalar early-warning score that crosses its decision boundary a median of 50.9 minutes before annotated onset across eleven calibrated test patients. Deployed on a physical Apple Watch, the model runs in a measured median of $156\mu s$ per window on the Neural Engine and compresses to a 33 KB package at 95.4% accuracy, making both detection and early warning feasible on resource-constrained wearable hardware.

Index Terms—atrial fibrillation, onset prediction, contrastive learning, RR intervals, wearable monitoring, edge inference, patient-aware representation learning

I. INTRODUCTION

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia, affecting over 37 million people worldwide and carrying a roughly fivefold elevated risk of ischaemic stroke [1], [2]. Its paroxysmal form (PAF), in which episodes self-terminate within seven days, is the earliest clinically actionable stage: reliable detection there opens a window for pharmacological and lifestyle intervention before progression to persistent forms [3], [4]. The clinical difficulty is that PAF is episodic and often asymptomatic. About one third of monitored patients remain symptom-free [5], so incidental detection

during a routine clinic visit is unreliable and continuous, long-term ambulatory monitoring becomes essential.

Modern smartwatches and single-lead patches detect cardiac R-peaks natively and stream RR-interval (RRI) sequences at negligible power cost, without storing or transmitting a full electrocardiogram (ECG) waveform [6], [7]. The same interval series can be recovered from photoplethysmography (PPG), the optical pulse signal ubiquitous in consumer wearables [8], [9], so an RRI-based model is signal-agnostic at the sensor. Because RRI streams are compact and are sensitive to the autonomic and electrophysiological changes that precede an episode [10]–[12], a small model can run directly on-device, enabling private, real-time screening. In resource-constrained settings where clinical-grade 12-lead monitoring is scarce, this offers a realistic path to early AF detection at scale.

Two obstacles stand between this promise and a deployable system. First, *generalization to unseen patients*: because every person carries a distinct heart-rhythm signature, supervised detectors trained on RRI dynamics often degrade sharply on individuals unlike the training population—Andersen et al. [13] report sensitivity falling from 98.98% under cross-validation to 86.04% on held-out patients. Second, *anticipation rather than mere detection*: a screening tool is far more useful if it flags an impending episode ahead of time, yet prior AF-prediction work relies on hand-crafted precursors and patient-mixed evaluation [14]–[17], leaving open whether one learned representation can serve both detection and forward-looking early warning under a strict patient-independent protocol.

This paper addresses both obstacles with one representation. Our contributions are:

- A **patient-aware contrastive objective** that restricts positive pairs to same-patient, same-class segments, preserving each patient’s individual sinus-rhythm (SR) manifold instead of collapsing it into a shared per-class prototype, paired with a compact multi-branch convolutional encoder suited to edge hardware.

- A **per-patient early-warning scheme** that reuses the frozen detection embedding, with no retraining, to produce a scalar onset score that crosses its decision boundary a median of 50.9 minutes before annotated AF onset.
- A **real on-device deployment** on a physical Apple Watch, with measured Neural-Engine latency of $\approx 156 \mu\text{s}$ per window and a compression study reaching a 33 KB model at 95.4% accuracy.

II. RELATED WORK

Detection from RR intervals. Early automated PAF analysis relied on hand-crafted heart-rate-variability (HRV) features—time-domain statistics, spectral power, and nonlinear measures such as sample entropy and the Poincaré plot [18]–[21]. Recurrent and convolutional networks have since superseded these: Faust et al. [22] report 98.51% accuracy with an LSTM, and Andersen et al. [13] a CNN–RNN with 98.98% sensitivity under cross-validation, though the same authors’ sensitivity drops by over 12 points on held-out patients—which motivates a patient-independent protocol as the honest measure of transfer.

Onset prediction. A separate line predicts PAF from the minutes preceding an episode, typically from atrial premature complexes or nonlinear HRV markers [14]–[16], surveyed in [17]. These forecasters are usually trained with within-patient data mixing and hand-designed precursors, leaving unclear whether a representation learned for detection can itself carry a forward-looking signal.

Contrastive representation learning. Contrastive learning is now the default for weakly-labeled physiological signals [23]–[26], with growing use in the ECG domain [27]–[29]. Supervised contrastive learning [25] pulls all same-class embeddings together, which on cardiac signals treats SR segments from all patients as interchangeable and discards the inter-patient structure that governs transfer. Patient-level variants such as CLOCS [30], PCLR [31], and a patient memory queue [32] use subject identity to form positives but remain *unsupervised*. Our objective is *supervised* and takes the intersection of class label and patient identity, so it neither blends the two classes nor collapses each patient into a shared prototype.

III. METHODOLOGY

A. Dataset and Preprocessing

We use IRIDIA-AF [33], comprising long-term single-lead ECG recordings from 167 patients with paroxysmal AF. An episode is retained when the AF duration is ≥ 1 hr and the immediately preceding SR duration is ≥ 4 hr. These thresholds are deliberate design choices: the ≥ 4 hr preceding-SR margin keeps the negative class clear of pre-onset transitional rhythm, since autonomic changes are known to emerge in the minutes to hours before an episode [10], while the ≥ 1 hr AF requirement ensures each positive segment is sustained and unambiguous. Splits are constructed at the patient level to prevent leakage (119/24/24 train/validation/test patients); after quality filtering 154/27/24 episodes remain.

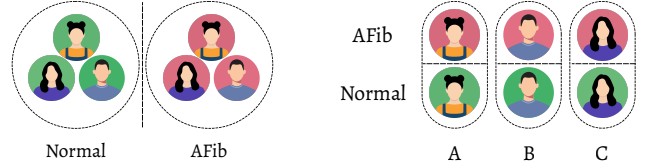


Fig. 1. Standard supervised contrastive learning (left) treats all same-class segments as positives regardless of patient. The proposed patient-aware formulation (right) restricts positives to same-patient, same-class segments, preserving each patient’s own SR baseline.

A RobustScaler removes each patient’s baseline heart rate,

$$\tilde{RR}_i = \frac{RR_i - \text{median}(\mathcal{R}_{\text{scale}}^{(p)})}{\text{IQR}(\mathcal{R}_{\text{scale}}^{(p)})}, \quad (1)$$

fit on the first, most temporally stable SR hour of patient p . Classification windows are drawn from a strictly disjoint subsequent hour, so the scaler never sees labeled data, and the ≥ 4 hr SR criterion keeps SR windows at least 2 hr clear of AF onset. The normalized stream is segmented with a sliding window of $W = 200$ beats and stride $S = 50$; beats shorter than 200 ms or longer than 2000 ms are discarded as physiologically implausible.

B. Patient-Aware Mini-Batch Sampling

Each mini-batch samples P patients, each contributing n SR and n AF windows, for a batch size of $B = 2nP$. A patient is eligible only if both its SR and AF pools hold at least n windows. This guarantees intra-patient, intra-class positive pairs in every batch, enforces class balance across patients, and prevents high-episode patients from dominating the training signal.

C. Patient-Aware Contrastive Objective

For an anchor i with class label y_i and patient identifier p_i , the positive and negative sets are

$$\mathcal{P}(i) = \{j \neq i \mid p_j = p_i \wedge y_j = y_i\}, \quad (2)$$

$$\mathcal{N}(i) = \{j \neq i \mid y_j \neq y_i\}. \quad (3)$$

Critically, same-class segments from *different* patients are excluded from $\mathcal{P}(i)$, which prevents the encoder from collapsing distinct individual SR baselines into a single shared prototype—the key departure from supervised contrastive learning [25] (Fig. 1). The per-anchor loss is the InfoNCE form

$$\mathcal{L}_i = -\log \frac{\sum_{j \in \mathcal{P}(i)} \exp(s_{ij})}{\sum_{j \in \mathcal{P}(i)} \exp(s_{ij}) + \sum_{k \in \mathcal{N}(i)} \exp(s_{ik})}, \quad (4)$$

with $s_{ij} = \hat{z}_i^\top \hat{z}_j / \tau$ and a learnable temperature τ ; the batch loss is $\mathcal{L} = \frac{1}{B} \sum_i \mathcal{L}_i$.

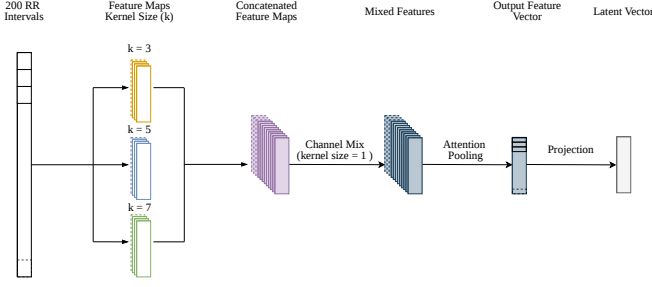


Fig. 2. Encoder: a multi-branch 1D-CNN backbone with temporal attention pooling followed by a two-layer MLP projection head. It is kept small so that any gain traces to the objective rather than to capacity, and so the footprint fits wearable hardware.

D. A Compact Encoder for the Edge

The encoder (Fig. 2) stays deliberately small. Three parallel 1D-CNN branches with kernel sizes $k \in \{3, 5, 7\}$ capture beat-to-beat fluctuations, medium-range oscillations, and broader trend dynamics, motivated by Inception-style multi-scale processing [34]. Each branch applies three strided convolution blocks (stride 2) with channel depths $\{16, 32, 64\}$, Group Normalization [35] (8 groups), and ReLU. Branch outputs are fused by a 1×1 convolution and pooled by a softmax temporal attention module [36],

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \mathbf{h}_t)}{\sum_{t'} \exp(\mathbf{w}^\top \mathbf{h}_{t'})}, \quad \mathbf{z} = \sum_t \alpha_t \mathbf{h}_t, \quad (5)$$

which up-weights physiologically salient segments without manual annotation. Following SimCLR [23], a two-layer multilayer perceptron (MLP) maps the pooled latent $\mathbf{z} \in \mathbb{R}^{128}$ to an embedding $\mathbf{e} \in \mathbb{R}^{128}$; both are ℓ_2 -normalized to the unit hypersphere [26]. The contrastive loss acts on $\hat{\mathbf{e}}$; downstream evaluation uses the pooled output $\hat{\mathbf{z}}$.

Training. PyTorch on a single NVIDIA RTX 2080. AdamW (learning rate 6.8×10^{-3} , weight decay 8.8×10^{-4}), cosine annealing to 10^{-6} , dropout 0.12 on the projection head, and a learnable temperature τ initialized at 0.05. Patient-aware sampling uses $P = 8$ patients and $n = 16$ windows per class ($B = 128$), for up to 100 epochs with early stopping on validation AUROC (patience 10). Thirty-two hyperparameters (architecture, optimizer, and window settings W, S) are tuned by Optuna’s TPE sampler [37] on the validation split, with the test split held out until final evaluation.

E. Per-Patient Early-Warning Score

We reuse the frozen detection representation, untouched, for a forward-looking early-warning scheme that reads directly off the geometry of the embedding space. For a given patient we embed one labeled SR episode and one labeled AF episode and collapse each to its class centroid, $\bar{\mu}_{\text{SR}}$ and $\bar{\mu}_{\text{AF}}$. These fix a personal *SR-to-AF axis* with unit direction

$$\hat{\mathbf{u}} = \frac{\bar{\mu}_{\text{AF}} - \bar{\mu}_{\text{SR}}}{\|\bar{\mu}_{\text{AF}} - \bar{\mu}_{\text{SR}}\|}. \quad (6)$$

Latent embedding and scalar projection along the SR-AF axis

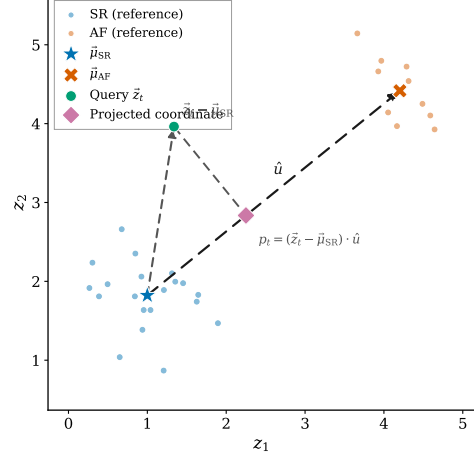


Fig. 3. Geometry of the early-warning scheme (illustrative 2-D sketch). A patient’s labeled SR and AF reference embeddings collapse to centroids $\bar{\mu}_{\text{SR}}$ and $\bar{\mu}_{\text{AF}}$, fixing the SR→AF axis $\hat{\mathbf{u}}$. A query window $\bar{\mathbf{z}}_t$ is referenced to $\bar{\mu}_{\text{SR}}$ and projected onto the axis; the score p_t is its normalized position.

Each incoming window is embedded to $\bar{\mathbf{z}}_t$, referenced to the SR centroid, projected onto the axis, and normalized by the axis length to give a scalar early-warning score

$$p_t = \frac{(\bar{\mathbf{z}}_t - \bar{\mu}_{\text{SR}}) \cdot \hat{\mathbf{u}}}{\|\bar{\mu}_{\text{AF}} - \bar{\mu}_{\text{SR}}\|}, \quad (7)$$

so that $p_t = 0$ sits at the SR centroid, $p_t = 1$ at the AF centroid, and $p_t = 0.5$ marks the boundary equidistant between them (Fig. 3). A climb of p_t across the midpoint toward the AF centroid flags an impending episode. The scheme asks only for a lightweight, few-shot calibration per patient, with *no retraining*. It holds together precisely because the patient-aware objective keeps each patient’s SR cluster tight and consistently placed, leaning on the very per-patient SR cohesion the objective was designed to produce.

IV. EXPERIMENTS AND RESULTS

Protocol. We adopt the standard contrastive evaluation: the encoder is frozen and a logistic-regression linear probe is fit on training-set embeddings, then evaluated on the held-out test split of entirely unseen patients. This measures representation quality independently of classifier capacity. Experiments are repeated over five independent seeds (controlling weight initialization and batch ordering on a fixed patient split) and reported as mean $\pm$ standard deviation.

A. Detection Performance

Table I reports per-class test performance on 4160 AF and 3799 SR segments from unseen patients. The proposed representation reaches an AUROC of 0.989 ± 0.003 and accuracy 0.952 ± 0.011 . AF recall (0.974 ± 0.008) consistently exceeds SR recall, a clinically favorable asymmetry that minimizes missed AF at the cost of a modest false-positive rate.

The advantage over baselines is chiefly one of *stability*. Holding the encoder, sampler, and probe fixed and varying

TABLE I
PER-CLASS LINEAR-PROBE PERFORMANCE ON THE
PATIENT-INDEPENDENT TEST SPLIT (4160 AF, 3799 SR SEGMENTS FROM
UNSEEN PATIENTS; MEAN $\pm$ STD OVER 5 SEEDS). BOLD MARKS THE
PRIMARY METRICS.

Class	Prec.	Recall	F1
SR (0)	0.970 $\pm$ 0.009	0.927 $\pm$ 0.020	0.948 $\pm$ 0.012
AF (1)	0.936 $\pm$ 0.017	0.974 $\pm$ 0.008	0.955 $\pm$ 0.010
Accuracy	0.952 $\pm$ 0.011		
AUROC	0.989 $\pm$ 0.003		

TABLE II
COMPARISON WITH PRIOR RRI-BASED AF DETECTION. DIRECT
NUMERICAL COMPARISON IS LIMITED BY DATASET AND PROTOCOL
DIFFERENCES. $\dagger$ CROSS-VALIDATION WITH WITHIN-PATIENT MIXING
INFLATES METRICS.

Study	Method	Validation	Sens.
Udawat et al. [18]	HRV+ML	CV $\dagger$	95.16%
Andersen et al. [13]	CNN+RNN	5-fold CV $\dagger$	98.98%
Andersen et al. [13]	CNN+RNN	Pt. holdout	86.04%
Faust et al. [22]	LSTM	10-fold CV $\dagger$	—
Ours	CNN+MLP	Pt. holdout	97.40 $\pm$ 0.80%

only the loss, the patient-aware objective attains AUROC 0.989 ± 0.003 against 0.983 ± 0.009 for supervised contrastive learning (SupCon) and 0.980 ± 0.012 for binary cross-entropy (BCE), reducing seed-to-seed AUROC variance by $2.6\times$ and $3.4\times$ respectively. This traces to embedding geometry: the objective attains the highest per-patient SR cohesion (0.850 vs. 0.800 SupCon, 0.772 BCE), whereas BCE produces the cleanest global separation yet the most disordered per-patient structure and the weakest transfer—confirming that per-subject consistency, not global separability, governs cross-patient generalization. Against prior RRI-based detectors under patient holdout (Table II), the representation attains $97.40 \pm 0.80\%$ sensitivity, over 11 points above Andersen et al.’s held-out sensitivity [13]. Applied zero-shot to the external SHDB-AF cohort [38] without any retraining, the same model retains an AUROC of 0.955, indicating transfer across acquisition device and population.

B. Anticipating AF Onset

We instantiate the early-warning scheme of Section III-E on the test patients. For each patient we fix a personal SR-to-AF axis (Eq. (6)) from a lightweight calibration drawn from that patient’s own labeled SR and AF windows, embed each window of a separate continuous pre-onset recording, and track the score p_t (Eq. (7)) across the hour before annotated onset, recording the earliest crossing of the midpoint $p_t = 0.5$ as the lead time.

Across the eleven test patients for whom both calibration windows and a continuous pre-onset recording are available, the early-warning score crosses the midpoint a median of 50.9 minutes (mean 39.1 ± 21.3 minutes) ahead of annotated

TABLE III
EARLY-WARNING LEAD TIME (MINUTES BEFORE ANNOTATED ONSET)
ACROSS THE ELEVEN CALIBRATED TEST PATIENTS WITH A CONTINUOUS
PRE-ONSET RECORDING. LEAD TIME IS THE EARLIEST MIDPOINT
CROSSING ($p_t = 0.5$) OF THE SCORE.

Statistic	Lead time (min)
Median	50.9
Mean $\pm$ std	39.1 ± 21.3
Range	3.0 – 59.0
Patients > 50 min ahead	6/11
Patients $\geq$ 10 min ahead	10/11

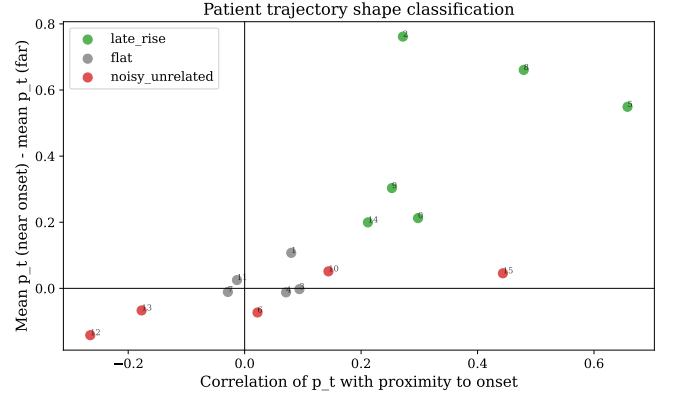


Fig. 4. Per-patient early-warning trajectory shapes. Across the sixteen calibrated test patients the p_t trajectories partition cleanly into six *late-rise* (score climbs as onset nears), five *flat*, and five *noisy* cases rather than a borderline continuum.

onset, ranging from 3.0 to 59.0 minutes (Table III). For six of the eleven patients the crossing occurs more than fifty minutes in advance, and for ten of eleven it occurs at least ten minutes ahead. Several lead times cluster near the 59-minute ceiling of the monitoring window, so those values are right-censored: the score has already reached the AF-ward half of the axis when monitoring begins, meaning the true lead time is longer than recorded. This is a proof-of-concept feasibility result obtained without any retraining, reusing the detection embedding as-is.

Per-patient heterogeneity. The population median conceals a wide spread. Across the sixteen test patients with enough labeled windows to fix a personal axis, the p_t trajectories sort into three clean shapes (Fig. 4): six *late-rise* (score climbs as onset nears), five *flat*, and five *noisy*. Two structural factors explain most of this. Reliability is gated by how far apart the SR and AF centroids sit: for three of sixteen the separation $\|\vec{\mu}_{AF} - \vec{\mu}_{SR}\|$ falls below 0.4 and the short axis amplifies embedding noise, whereas the thirteen above 0.7 give stable trajectories. And calibration is only as good as its reference episodes: a contaminated SR reference drags the centroid off position. These are calibration-quality failures, not failures of the score itself, and point to a per-patient calibration screen as a prerequisite for deployment.

TABLE IV
MEASURED ON-DEVICE COMPRESSION TRADE-OFF (DEPLOYED CLASSIFIER, IRIDIA-AF PATIENT-DISJOINT TEST SET). `float16` IS THE SHIPPED VARIANT; PTQ DENOTES POST-TRAINING QUANTIZATION AND QAT QUANTIZATION-AWARE TRAINING.

Variant	Size	Compr.	Acc. %	AUROC
<code>float32</code> baseline	462 KB	1.0×	95.84	0.987
<code>float16</code> (shipped)	261 KB	2.0×	95.86	0.987
<code>int8</code> (PTQ)	164 KB	3.7×	95.84	0.987
palettize 4-bit	111 KB	7.0×	95.76	0.987
palettize 2-bit (QAT)	33 KB	12.6×	95.40	0.985

C. On-Device Deployment

To confirm that the approach runs at the edge rather than only in principle, we deployed the model on a physical Apple Watch. For deployment we collapse the frozen-encoder-plus-probe pipeline into a single network: the same multi-scale backbone, fine-tuned with a compact classification head (≈ 106 K parameters), which runs as one forward pass and reaches accuracy consistent with the linear probe (95.84%, Table IV). It is exported to CoreML (mlprogram, `float16`, watchOS 10), consuming 200 RR intervals per window and emitting a single AF probability. We profiled it on-device with Instruments (CoreML, Neural Engine, and energy signposts) over a 61-second stress run of 831 inferences.

Inference is effectively free on this hardware. Median execution on the Apple Neural Engine is $155.7 \mu\text{s}$ (p95 $229 \mu\text{s}$) at a duty cycle of 0.21%, and the `float16` package occupies about 262 KB on disk. Because the model is so small, latency is compute-unit-insensitive: a forward pass is sub-millisecond on the CPU as well, so the Neural Engine’s value is energy, not speed. In normal use the app is gated by data collection rather than compute: a first prediction needs 200 intervals (minutes of wear) and then re-fires about once per minute, so the dominant power cost is the always-on heart-rate sensor, not the network.

The footprint can be cut much further with training-time compression (Table IV). Post-training `int8` quantization is near-lossless (3.7× smaller, accuracy unchanged), and palettization-aware fine-tuning reaches 12.6× compression at 2-bit: a 33 KB model that retains 95.4% accuracy and an AUROC of 0.985. Post-training magnitude pruning, by contrast, collapses without fine-tuning, and sensitivity stays robust while specificity is the fragile metric under every compression scheme.

Two properties of the encoder further aid streaming inference: Group Normalization uses no batch statistics and so behaves identically at batch size one, and strided convolutions shrink the sequence early, keeping the deeper layers cheap. The early-warning score of Section III-E adds only two centroid lookups and a dot product on top of one encoder pass, so anticipation is essentially free once detection runs.

V. DISCUSSION

The two results reinforce one mechanism. Restricting positives to same-patient, same-class segments keeps each in-

dividual’s SR cluster tight and consistently placed on the hypersphere; a t-SNE projection of the frozen embeddings shows that, beyond the global SR/AF split, segments retain visible per-patient structure rather than collapsing into a single per-class prototype. This is what stabilizes cross-patient detection (the $2.6\times$ variance reduction over SupCon) and is also what makes the early-warning axis well-defined per patient: a consistently placed SR centroid gives a reliable reference direction to project against. Anticipation therefore comes for free from a representation built for detection, with no dedicated forecaster or retraining. Clinically, the consistently high AF recall (0.974 ± 0.008) is what matters: a missed episode may delay anticoagulation and elevate stroke risk, so seed-stable sensitivity is a genuine deployment advantage.

Limitations. Three caveats bound these claims. The early-warning analysis is a feasibility study on eleven calibrated patients with right-censored lead times, not a validated forecaster; the clean partition of trajectories into late-rise, flat, and noisy cases shows that per-patient calibration quality must be screened before deployment. Detection is validated on a single dataset (with one external zero-shot cohort), so multi-centre validation remains open. Finally, the deployed watch feeds BPM-derived intervals (60000/bpm) rather than true beat-to-beat R-R timing, and since AF’s signature is beat-to-beat irregularity, on-device predictive accuracy may differ from the IRIDIA-AF test-set figures even though runtime performance is excellent. Converting the feasibility signal into a calibrated forecaster and validating on consumer-grade PPG waveforms are the natural next steps.

VI. CONCLUSION

We presented a single patient-aware representation that serves both AF detection and forward-looking early warning from RR-interval streams. By restricting contrastive positives to same-patient, same-class segments, the encoder preserves each patient’s own sinus-rhythm baseline instead of collapsing it into a shared prototype, reaching a patient-independent AUROC of 0.989 ± 0.003 with $2.6\times$ lower seed variance than supervised contrastive learning. Reusing the frozen embedding with only a lightweight per-patient calibration, a sinus-rhythm-to-AF projection anticipates onset a median of 50.9 minutes ahead across eleven test patients, at essentially no additional cost. Deployed on a physical Apple Watch, the model runs in a measured $156 \mu\text{s}$ per window and compresses to 33 KB, placing both detection and anticipation within reach of resource-constrained wearable hardware.

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REFERENCES

- [1] J. A. Joglar *et al.*, “2023 ACC/AHA/ACCP/HRS Guideline for the Diagnosis and Management of Atrial Fibrillation: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines,” *Circulation*, vol. 149, no. 1, pp. e1–e156, Jan. 2024.
- [2] M. K. Chung *et al.*, “Atrial fibrillation,” vol. 75, no. 14, pp. 1689–1713, publisher: American College of Cardiology Foundation. [Online]. Available: <https://www.jacc.org/doi/10.1016/j.jacc.2020.02.025>
- [3] G. Hindricks *et al.*, “2020 esc guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the european association for cardio-thoracic surgery (eacts).” *European Heart Journal*, vol. 42, no. 5, pp. 373–498, 2020, mAG ID: 3082188176.
- [4] M. Vinciguerra, D. Dobrev, and S. Nattel, “Atrial fibrillation: pathophysiology, genetic and epigenetic mechanisms,” vol. 37, p. 100785. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S2666776223002041>
- [5] R. R. De With *et al.*, “Temporal patterns and short-term progression of paroxysmal atrial fibrillation: data from RACE V,” *Europace*, vol. 22, no. 8, pp. 1162–1172, Aug. 2020.
- [6] M. V. Perez *et al.*, “Large-scale assessment of a smartwatch to identify atrial fibrillation,” *New England Journal of Medicine*, vol. 381, no. 20, pp. 1909–1917, 2019.
- [7] J. Wasserlauf, C. You, R. Patel, A. Valys, D. Albert, and R. Passman, “Smartwatch performance for the detection and quantification of atrial fibrillation,” *Circulation: Arrhythmia and Electrophysiology*, vol. 12, p. e006834, 2019.
- [8] Y. Guo *et al.*, “Photoplethysmography-based machine learning approaches for atrial fibrillation prediction: A report from the huawei heart study,” *JACC: Asia*, vol. 1, no. 3, pp. 399–408, 2021. [Online]. Available: <https://doi.org/10.1016/j.jacasi.2021.09.004>
- [9] K. H. Shelley, “Photoplethysmography: Beyond the calculation of arterial oxygen saturation and heart rate,” *Anesthesia & Analgesia*, vol. 105, no. 6, pp. S31–S36, December 2007.
- [10] J.-M. Gregoire, C. Gilon, F. Marelli, P. Godart, H. Bersini, and S. Carlier, “Autonomic nervous system activity before atrial fibrillation onset as assessed by heart rate variability,” *Rev Cardiovasc Med*, vol. 26, no. 1, p. 25364, Jan. 2025.
- [11] J.-M. Gregoire, C. Gilon, S. Carlier, and H. Bersini, “Role of the autonomic nervous system and premature atrial contractions in short-term paroxysmal atrial fibrillation forecasting: Insights from machine learning models,” *Archives of Cardiovascular Diseases*, vol. 115, no. 6, pp. 377–387, Jun. 2022.
- [12] Z. Yin *et al.*, “Identification of atrial fibrillation using heart rate variability: a meta-analysis,” *Frontiers in Cardiovascular Medicine*, vol. 12, p. 1581683, 2025.
- [13] R. S. Andersen, A. Peimankar, and S. Puthusserypady, “A deep learning approach for real-time detection of atrial fibrillation,” *Expert Systems with Applications*, vol. 115, pp. 465–473, Jan. 2019.
- [14] T. Thong, J. McNames, M. Aboy, and B. Goldstein, “Prediction of paroxysmal atrial fibrillation by analysis of atrial premature complexes,” *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 4, pp. 561–569, 2004.
- [15] M. Mohebbi and H. Ghassemian, “Prediction of paroxysmal atrial fibrillation based on non-linear analysis and spectrum and bispectrum features of the heart rate variability signal,” *Computer Methods and Programs in Biomedicine*, vol. 105, pp. 40–49, 2012.
- [16] E. Ebrahimzadeh, M. Kalantari, M. Joulani, R. S. Shahraki, F. Fayaz, and F. Ahmadi, “Prediction of paroxysmal atrial fibrillation: A machine learning based approach using combined feature vector and mixture of expert classification on hrv signal,” *Computer Methods and Programs in Biomedicine*, vol. 165, pp. 53–67, 2018.
- [17] A. S. Tseng and P. A. Noseworthy, “Prediction of atrial fibrillation using machine learning: A review,” *Frontiers in Physiology*, vol. 12, p. 752317, 2021.
- [18] A. S. Udawat and P. Singh, “An automated detection of atrial fibrillation from single-lead ECG using HRV features and machine learning,” *J Electrocardiol*, vol. 75, pp. 70–81, 2022.
- [19] Y. V. Chesnokov, “Complexity and spectral analysis of the heart rate variability dynamics for distant prediction of paroxysmal atrial fibrillation with artificial intelligence methods,” *Artif Intell Med*, vol. 43, no. 2, pp. 151–165, Jun. 2008.
- [20] J. Saiz-Vivo, V. D. A. Corino, R. Hatala, M. de Melis, and L. T. Mainardi, “Heart Rate Variability and Clinical Features as Predictors of Atrial Fibrillation Recurrence After Catheter Ablation: A Pilot Study,” *Front. Physiol.*, vol. 12, May 2021.
- [21] R. Rouhi, M. Clausel, J. Oster, and F. Lauer, “An Interpretable Hand-Crafted Feature-Based Model for Atrial Fibrillation Detection,” *Front Physiol*, vol. 12, p. 657304, May 2021.
- [22] O. Faust, A. Shenfield, M. Kareem, T. R. San, H. Fujita, and U. R. Acharya, “Automated detection of atrial fibrillation using long short-term memory network with RR interval signals,” *Comput Biol Med*, vol. 102, pp. 327–335, Nov. 2018.
- [23] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *Proceedings of the 37th International Conference on Machine Learning*, ser. ICML’20, vol. 119. JMLR.org, Jul. 2020, pp. 1597–1607.
- [24] J.-B. Grill *et al.*, “Bootstrap Your Own Latent - A New Approach to Self-Supervised Learning,” in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 21 271–21 284.
- [25] P. Khosla *et al.*, “Supervised Contrastive Learning,” Mar. 2021, arXiv:2004.11362 [cs].
- [26] T. Wang and P. Isola, “Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere,” in *Proceedings of the 37th International Conference on Machine Learning*. PMLR, Nov. 2020, pp. 9929–9939.
- [27] W. Liu *et al.*, “Dense lead contrast for self-supervised representation learning of multilead electrocardiograms,” *Information Sciences*, vol. 634, pp. 189–205, Jul. 2023.
- [28] J. Hu, C. Li, J. Cao, and B. Kou, “A novel multimodal self-supervised framework for ECG arrhythmia classification,” *Computers in Biology and Medicine*, vol. 198, p. 111137, Nov. 2025.
- [29] W. Chen, H. Wang, L. Zhang, and M. Zhang, “Temporal and spatial self supervised learning methods for electrocardiograms,” *Sci Rep*, vol. 15, no. 1, p. 6029, Feb. 2025.
- [30] D. Kiyasseh, T. Zhu, and D. A. Clifton, “CLOCS: Contrastive Learning of Cardiac Signals Across Space, Time, and Patients,” in *Proceedings of the 38th International Conference on Machine Learning*. PMLR, Jul. 2021, pp. 5606–5615.
- [31] N. Diamant, E. Reinertsen, S. Song, A. D. Aguirre, C. M. Stultz, and P. Batra, “Patient contrastive learning: A performant, expressive, and practical approach to electrocardiogram modeling,” *PLoS Comput Biol*, vol. 18, no. 2, p. e1009862, Feb. 2022.
- [32] X. Sun, Y. Yang, and X. Dong, “Enhancing Contrastive Learning-based Electrocardiogram Pretrained Model with Patient Memory Queue,” May 2025, arXiv:2506.06310 [eess].
- [33] C. Gilon, J.-M. Grégoire, M. Mathieu, S. Carlier, and H. Bersini, “IRIDIA-AF, a large paroxysmal atrial fibrillation long-term electrocardiogram monitoring database,” *Sci Data*, vol. 10, no. 1, p. 714, Oct. 2023.
- [34] C. Szegedy *et al.*, “Going deeper with convolutions,” in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2015, pp. 1–9, iSSN: 1063-6919.
- [35] Y. Wu and K. He, “Group normalization,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, September 2018.
- [36] S. P. Shashikumar, A. J. Shah, G. D. Clifford, and S. Nemat, “Detection of Paroxysmal Atrial Fibrillation using Attention-based Bidirectional Recurrent Neural Networks,” in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, ser. KDD ’18. New York, NY, USA: Association for Computing Machinery, Jul. 2018, pp. 715–723.
- [37] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A Next-generation Hyperparameter Optimization Framework,” Jul. 2019, arXiv:1907.10902 [cs].
- [38] K. Tsutsui, S. Biton Brimer, and J. Behar, “SHDB-AF: a Japanese Holter ECG database of atrial fibrillation,” *PhysioNet*, Apr. 2025, version 1.0.1.